



Managing Resilient Workforces in Logistics: The Need for Psychological Well-Being for Drivers

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Abstract: The psychological well-being of truck drivers is a distinct concern in occupational psychology and engineering management. Job isolation, work insecurity, and poor working conditions are the factors that contribute to elevated stress and turnover among truck drivers. However, limited studies could be found with these factors that ultimately influence truck drivers' psychological well-being. Underpinned by balance theory, this study examines the relationships between working environment, social support, economic domains and truck drivers' psychological well-being, with the mediating effect of quality of life (QoL). A primary quantitative approach was administered to 403 truck drivers. The data were then analyzed using partial least squares–structural equation modeling (PLS–SEM) approach and mediation analysis with bootstrapping procedures. The results indicate that working environment, social support, and economic domains significantly influenced overall QoL. Crucially, overall QoL fully mediated the relationships between working environment, social support, economic conditions, and psychological well-being. However, the direct paths from these occupational factors to psychological well-being were statistically non-significant. This research would strengthen occupational psychology and engineering management fields theoretically as overall QoL is used as a key mediator between occupational stressors and psychological well-being of the truck drivers. These findings may help logistics managers identify relevant areas for economic, social, and organizational intervention. Practically, these results emphasize the need for holistic interventions to create fair compensation and ergonomic work design to socially support and enhance truck drivers' well-being.

Keywords: Truck drivers; Quality of life; Psychological well-being; Social support; Working environment

1 Introduction

Occupational transportation safety is an important public priority that requires collective commitment from all stakeholders [1]. One aspect that is often overlooked is the influence of a truck driver's psychological well-being on their behavior behind the wheel. A prior study [2] indicates that 44% of adults concerned about the psychological health issues have emerged as the most significant healthcare concern globally and in Malaysia. Truck drivers have always been a vulnerable group for the country's gross domestic product; however, they face elevated psychological health issues due to occupational stressors. Furthermore, according to the study [3], psychological distress among Australian truck drivers are one of the main tragic outcomes from neglected psychological health. Such a "tough" occupational environment among Malaysian truck drivers often clashes with the poor ergonomic working conditions and low wages [4]. Truck drivers may experience a demanding and stressful working environment [5]. An effective workplace environment with adequate organizational support are vital factors to improve workforce morale and reduce operational resilience [6]. This highlights the vital role that structural empowerment criteria play in safeguarding employee performance.

Logistics transportation serves as a vulnerable operational node in the discipline of engineering management [7]. In road freight transport, truck drivers play a critical role in sustaining transportation system stability [8]. The reliance on stimulants and long working hours may operationally affect the Malaysian truck drivers' psychological factors [9]. The cultural values within the Malaysian logistics industry encourage resilience and stoicism, which potentially worsen the occupational safety and psychological distress among truck drivers. While operational psychological

health among truck drivers is a globally recognized concern, the Malaysian logistics sector presents a unique and high-pressure empirical context [10]. Malaysia serves as a central logistical hub in Southeast Asia. However, the rapid growth of transportation sector has placed massive stress on its workforce. Malaysian Institute of Road Safety Research (MIROS) further highlights that truck drivers are a high-risk group associated with fatigue-related traffic accidents [11]. This is due to insufficient rest and relaxation facilities along major expressways and the nature of demanding schedules among the truck drivers. Moreover, the low psychological well-being of truck drivers often correlates with a higher likelihood of substance abuse. For example, truck drivers in Southeast Queensland often rely on drugs or illegal medications to stay alert and fight off fatigue, leading to the potential for drug abuse [12]. A previous study [9] also raised concerns that Malaysian truck drivers are chronically fatigue working under tight delivery schedules. Comparatively, a previous study of commercial drivers in Ghana [13] found that such working conditions could endanger drivers' health and lead to dangerous societal implications. This can introduce significant systemic risk when truck drivers' psychological well-being was compromised due to occupational stressors [14]. Consequently, drug misuse increases accident rates that would lead to injuries, fatalities, and economic losses in the transportation industry [15].

Hence, addressing these psychological issues is essential to improving the overall safety and efficiency of the industry. In addition, the factors such as poor working environment, intense workload, lack of management support, and financial insecurity are among the main factors affecting psychological health based on the study [16]. Besides, managing truck drivers' well-being would serve as a fundamental strategy for maintaining logistics system reliability and mitigating operational disruptions. This study focuses on understanding the occupational factors that contribute to the psychological well-being of drivers. Neglecting drivers' psychological health has implications beyond the individual level. It could potentially lead to consequences that can cause widespread societal harm. Thus, highlighting these issues would contribute to a better understanding of the occupational hazards faced by truck drivers aside from raising awareness of systemic changes that support the psychological well-being of this crucial workforce. Next section discusses the research model grounded in balance theory to further untangle these occupational stressors on its impact on the psychological well-being of drivers.

2 Literature Review

Psychological well-being is a multi-dimensional construct [17]. It includes self-acceptance, personal growth, purpose in life, environmental mastery, and positive relations with others [18]. Since truck drivers constantly face challenges such as safety concerns, long-haul isolation and irregular schedules, it is important to understand the truck drivers' psychological well-being in workplaces [19]. A previous study [20] explored how well-being affects drivers' resilience, coping strategies, and performance. The study [21] examined the relationship between job stress, psychological strain, and employee turnover. Besides, the study [16] examined the influences of stress and personality on fatigue. The study [22] suggests, on the other hand, psychological health directly associates with accident rates, raising the emphasis for the need of focused wellness initiatives. The study [3] highlights the need to design comprehensive psychological wellness programs tailored to the trucking industry. All these studies emphasize that the understanding of psychological well-being is vital to develop effective support and interventions among truck drivers. This further strengthens the association between truck drivers' psychological well-being and their occupational safety and productivity.

This research is underpinned by Fritz Heider's balance theory, which suggests that psychological equilibrium within interpersonal relationships is essential for psychological harmony [23]. Under this theory, the P-O-X model is posited with a triangular representation of the dynamics between persons (P), others (O), and context (X). In the context of this study, balance theory is applied through the P-O-X model, where P represents the truck driver, O represents the external environment (comprising the working environment, social support, and economic conditions), and X represents the truck drivers' quality of life (QoL). According to the study [24], individuals are motivated to maintain a state of cognitive consistency or balance. A balanced state exists when the relationships between these three entities—P, O and X are positive. Firstly, the truck drivers (P) inherently desire and value a high QoL (X) for themselves and their families. Second, the truck drivers (P) are strongly tied to their profession and external environment (O) because it is their livelihood. If the external environment or stressors (O) such as the poor working environment, lack of social support, or low wages negatively impact their QoL, the truck drivers' psychological well-being would be degraded. Thus, the two positive links and one negative link result in a state of cognitive imbalance as indicated in Figure 1. In the trucking industry, a state of cognitive imbalance occurs when the truck drivers perceive a negative relationship between poor ergonomic working environments or financial instability in economic domains. This imbalance would raise psychological tension and discomfort among the truck drivers. Psychological tension reflects the cognitive dissonance truck drivers experience when they struggle to reconcile their vital need to work with the fact that the work is actively degrading their QoL [20]. As the driver cannot easily change the economic domains or working environment (O), the unresolved tension manifests internally as burnout, stress, fatigue, and ultimately, degraded psychological well-being [25]. Thus, this research studies the relationship between

the environmental stressors truck drivers face (working environment), such as work-related stress, the intricate web of their interpersonal relations (social support), and the perception of inequitable compensation for extensive working hours (economic domains) from the perspective of truck drivers [26] underpinned by balance theory.

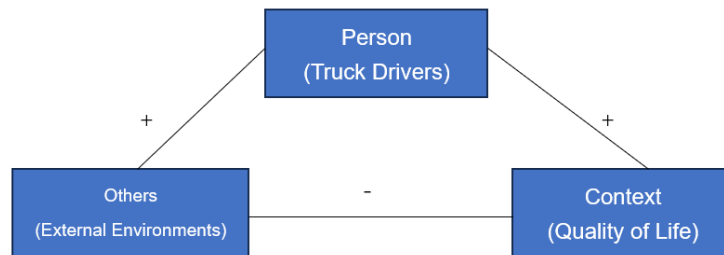


Figure 1. Triad relationship between persons (P), others (O), and context (X)

Source: Adapted from Ref. [23].

2.1 Working Environment

Working environment is defined as the physical and psychosocial factors that describe the job context. It refers to the workspace ergonomics, work culture, working hours, organizational and psychosocial job conditions [27]. Truck drivers who face irregular schedules are more susceptible to job dissatisfaction, stress, and potential psychological health issues [28, 29]. Furthermore, prior research has noted the detrimental effects of work exhaustion and social isolation in long-haul truck drivers, highlighting the urgent need for policies and occupational therapy programs aimed at improving truckers' psychological health [30]. Consequently, a conducive working environment would increase job satisfaction and improve health wellness of truck drivers. Based on the study [29], ergonomically designed vehicles and supportive company policies would further improve job satisfaction and reduce health issues. Besides, workplace intervention approaches such as addressing shift scheduling, trust between drivers and supervisors, and the physical working environment are crucial in improving drivers' health and safety. Thus, this study proposes that a favorable working environment that significantly influences drivers' QoL and psychological well-being. Accordingly, the following hypotheses are proposed:

H1: Working environment is positively associated with the overall QoL of drivers.

H8: Working environment is positively associated with the psychological well-being of drivers.

2.2 Social Support

Social support refers to the recognition of the employees from their employers or peer colleagues at the workplace. It measures how individuals feel valued, understood, and connected within their workplace environment [31–33]. Social support serves as a robust system to create healthier workspaces, mitigate stress, and reduce psychological health risks among employees [34]. Similarly, based on the research [20], truck drivers often lack social support due to their isolated working conditions. The study [35] also examined that drivers with robust social support, including emotional support, practical help, and community resources would generally have lower levels of job-related stress and improved psychological health outcomes. Besides, the study [36] explored how psychological health is intertwined with social capital. This underscores the strong relationship between psychological health outcomes and elements of social capital such as workplace support. This would further reinforce the importance of having supportive social networks in the workplace's psychological health. Additionally, the study [37] demonstrates that social support significantly correlates with seafarers' overall QoL. Hence, this research aims to deepen the understanding of how social dynamics within the workplace correlate with the psychological health of drivers, offering insights for future workplace policy and psychological well-being initiatives. This leads to the following hypotheses:

H2: Social support is positively associated with the overall QoL of drivers.

H9: Social support is positively associated with the psychological well-being of drivers.

2.3 Economic Domains

Economic domains refer to income level, job stability, and financial security. Firstly, income levels or salary would affect truck drivers' ability to meet personal and family needs. This factor would affect one's life satisfaction. Second, job stability would assure psychological comfort that reduce stress and anxiety among truck drives as they have more predictable income and employment continuity. Thirdly, financial security provides a safety net that enhances psychological comfort and stability against unexpected economic downturns. Thus, all these factors collectively affect the truck drivers' QoL and psychological health [38]. The study [39] further supports the critical

impact of economic factors on life satisfaction. Financial stability is associated with psychological health in the trucking industry. The study [40] states that unstable financial situations in employment often create psychological stress among workers. All these insights highlight the importance of examining the financial conditions of truck drivers to understand their impact on psychological health [41]. Therefore, an improved economic stability could have profound impact on the truck drivers' overall well-being. Based on these dynamics, it is hypothesized that:

- H3: Economic domains are positively associated with the overall QoL of truck drivers.
- H10: Economic domains are positively associated with the psychological well-being of drivers.

2.4 Quality of Life

QoL focuses on the interaction among interpersonal, physical, mental, and spiritual health and environmental conditions. QoL is particularly important in truck driver's occupational context because improved rest can positively affect their overall health [30]. QoL is defined as general life satisfaction and global subjective well-being [42] in this research context. Rather than measuring specific medical conditions or physical health, it captures the truck drivers' overall cognitive evaluation of their life circumstances and achievements. When truck drivers perceive that their working environment and economic domains allow them to attain a life close to their ideal, their general life satisfaction improves. This holistic view is further supported by the studies [41, 43], which is important to consider work-life balance and psychological health dimensions in assessing QoL. QoL can serve as a critical mediator to demonstrate that improvements in the working environment, social support, and economic domains could potentially elevate truck drivers' overall QoL. This further highlights the need for targeted interventions and policy initiatives to improve drivers' QoL and psychological well-being. Taken together, these studies suggest that occupational conditions may influence drivers' psychological well-being indirectly by shaping their perceived QoL. Thus, the following hypotheses are postulated:

- H4: Overall QoL is positively associated with the psychological well-being of drivers.
- H5: Overall QoL mediates the relationship between the working environment and the psychological well-being of drivers.
- H6: Overall QoL mediates the relationship between social support and the psychological well-being of drivers.
- H7: Overall QoL mediates the relationship between the economic domains and the psychological well-being of truck drivers.

This research focuses on how the overall QoL mediates the relationship between working environment, social support, economic domains, and psychological well-being of drivers, as indicated in Figure 2. By examining these dimensions, this study would provide valuable insights into potential areas for intervention and support that shape the psychological health and well-being for truck drivers.

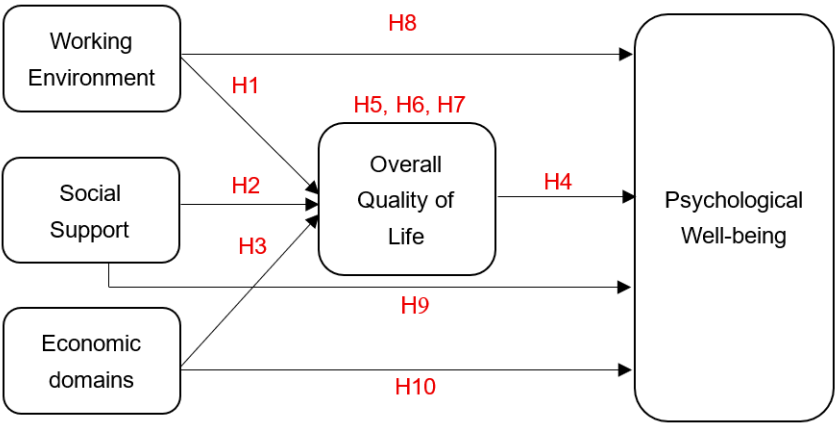


Figure 2. Research model

Note: H5, H6, and H7 refer to mediation effects.

Next section outlines the quantitative methodology used to empirically test the research model.

3 Methodology

This research employed a quantitative survey questionnaire for data collection using a convenience-sampling approach to test the hypotheses; the target population comprised truck drivers in Malaysia. This target group was chosen because it is recognized for its critical role in the regional domestic transportation economy and the unique challenges they face [44]. The data were collected through self-administered online questionnaires from November 2025 to January 2026. The sampling frame for company-employed drivers was obtained from official

industry bodies, primarily the Association of Malaysian Haulers (AMH) [45] and the Federation of Malaysian Freight Forwarders (FMFF) [46]. By utilizing the publicly available databases from AMH and FMFF, respondents were contacted through human resource departments and dispatch managers of registered logistics companies to assist in distributing the online survey links. These respondents were company-employed truck drivers, including a mix of long-haul drivers and short-haul drivers. The structure of the questionnaire was categorized into drivers' demographic profiles, followed by the operationalized constructs on working environment, social support, economic domains, overall QoL and psychological well-being. In this study, categorization of truck drivers' income was treated as an independent descriptive demographic variable. The monthly income categories are established purely as operational definitions for analytical purposes to profile the sampled workforce.

Survey items for working environment, social support and psychological well-being were adapted from the studies [38]. Working environment consists of five items with an example item "The management at my workplace treats me fairly". Social support consists of five items with a sample item "When I am in trouble, I tend to openly discuss my problems to seek advice and support". Psychological well-being consists of six items with the sample "I still enjoy the things I used to do". Survey items for economic domain and overall QoL were adapted from the study [42]. Economic domain consists of six items with an example item "My job provides well for my family's needs". Overall QoL consists of six items with a sample item "I am satisfied with my life as a whole". The credibility of the questionnaire was then pre-tested by industry practitioners and academicians, following the content validity procedures suggested by the study [47]. All the content validity indices showed a high level of acceptance with the values above 0.70 [47] for the factors affecting psychological well-being and overall QoL measurements. The frequency analysis was next performed to analyze the demographic data of 403 respondents by conducting a frequency analysis using the descriptive statistical method as shown in Table 1 below. Notably, Table 1 indicates that 25.6% of the respondents were female commercial drivers. Although the traditional heavy-haulage sector remains predominantly male-dominated [48], this higher proportion of female participation is attributed to the inclusive scope of the study. This encompasses truck drivers of light-to-medium trucks operating in last-mile e-commerce distribution and urban logistics hubs. Additionally, the digital distribution of the survey via logistics community networks may have yielded a higher response engagement among female operators.

Table 1. Demographic profiles of respondents (N = 403)

Demographic Items	Categories	Frequency	Percentage (%)
Age	<26	87	21.6
	26–35	118	29.3
	36–45	122	30.3
	46–55	63	15.6
	>55	13	3.2
Gender	Male	300	74.4
	Female	103	25.6
Level of education	Secondary education	232	57.6
	Certificate	27	6.7
	Diploma	70	17.4
	Degree	74	18.4
Driving experience (years)	<3	83	20.6
	3–5	114	28.3
	>5	206	51.1
Income	<RM2,500	102	25.3
	RM2,500–RM3,169	75	18.6
	RM3,170–RM3,969	47	11.7
	RM3,970–RM4,849	47	11.7
	RM4,850–RM5,879	66	16.4
	RM5,880–RM7,099	43	10.7
	>RM7,099	23	5.6

Note: RM = Ringgit Malaysia.

4 Findings

The research evaluated the measurement model and the structural model. The composite reliability, internal consistency reliability (α), and convergent validity are utilized to determine the reliability and validity of the outer measurement model [49]. Table 2 indicates that composite reliability values ranged from 0.843 to 0.960; α values ranged from 0.771 to 0.950, signifying acceptable internal reliability [50]. Following the partial least

squares–structural equation modeling (PLS–SEM) guidelines, the inner variance inflation factor values were assessed and all values were found below the conservative threshold of 3.3 [50]. Thus, this suggested that common method bias did not pose a significant threat to the results. Six items (WE2, SS3, QoL6, PWB2, PWB3, PWB6) were removed during the assessment of the measurement model to establish indicator reliability. The initial outer loadings for these items fell below the recommended threshold of 0.70 [50]; specifically: WE2 = 0.339, SS3 = 0.450, QoL6 = 0.386, PWB2 = 0.446, PWB3 = 0.348, and PWB6 = 0.470. Substantive analysis of these deleted items revealed context-specific reasons for their low performance. For example, item WE2 “I can talk to my immediate supervisor about difficulties I experience at work” may vary too widely for truck drivers as compared to office workers. These items failed to align with the unique operational realities of the truck-driving profession, which is characterized by extreme physical isolation, chronic fatigue, and susceptibility to external economic inflation. Crucially, the removal of these items did not compromise content validity. The remaining high-loading indicators firmly preserve the core theoretical dimensions of the constructs. Thereby, it ensures a more contextually accurate and statistically robust measurement model for the logistics workforce.

Table 2. Measurement model

Construct	Item	Loadings	α	Composite Reliability	Average Variance Extracted	Variance Inflation Factor
WE	WE1	0.923	0.855	0.902	0.699	2.315
	WE3	0.812				
	WE4	0.823				
	WE5	0.777				
SS	SS1	0.828	0.771	0.843	0.574	2.173
	SS2	0.747				
	SS4	0.703				
	SS5	0.746				
ED	ED1	0.915	0.950	0.960	0.802	1.651
	ED2	0.914				
	ED3	0.951				
	ED4	0.912				
	ED5	0.842				
	ED6	0.832				
Overall QoL	QoL1	0.744	0.897	0.925	0.713	1.000
	QoL2	0.909				
	QoL3	0.910				
	QoL4	0.868				
PWB	QoL5	0.777	0.765	0.865	0.681	2.489
	PWB1	0.818				
	PWB4	0.873				
	PWB5	0.783				

Note: WE2, SS3, QoL6, PWB2, PWB3, and PWB6 were deleted due to low loadings. WE = working environment; SS = social support; ED = economic domains; QoL = quality of life; PWB = psychological well-being.

Next, convergent validity is determined with the average variance extracted values above 0.5 [49]. Discriminant validity is assessed through the Fornell-Larcker criterion [50]. In relation to the study [49], the bolded diagonal square roots of average variance extracted values in Table 3 exceed the inter-variable correlation coefficients. Therefore, this indicates a strong discriminant validity.

Table 3. Fornell-Larcker criterion

	ED	PWB	QoL	SS	WE
ED	0.896				
PWB	0.497	0.825			
QoL	0.805	0.477	0.844		
SS	0.562	0.414	0.643	0.757	
WE	0.598	0.536	0.679	0.715	0.836

Note: WE = working environment; SS = social support; ED = economic domains; QoL = quality of life; PWB = psychological well-being.

4.1 Structural Model

Multivariate skewness and kurtosis were assessed using the guidelines from the study [51]. The data were not multivariate normal with Mardia's multivariate skewness at $\beta = 61.468$, $p < 0.01$ and Mardia's multivariate kurtosis at $\beta = 243.115$, $p < 0.01$. Thus, a 5,000 resampling bootstrapping procedure was performed to obtain the path coefficients, standard errors, t -values, and p -values of the structural model [49]. The results of hypothesis testing are shown in Table 4 and Table 5.

Table 4. Hypothesis testing (direct effect)

	Std. β	Std. Error	t -value	p -value	BCI Lower Limit	BCI Upper Limit	f^2
WE $\rightarrow$ QoL	0.212	0.047	4.487	0.000	0.130	0.287	0.070
SS $\rightarrow$ QoL	0.161	0.046	3.538	0.000	0.091	0.239	0.043
ED $\rightarrow$ QoL	0.587	0.029	20.213	0.000	0.538	0.635	0.747
QoL $\rightarrow$ PWB	0.477	0.037	12.762	0.000	0.418	0.541	0.295
WE $\rightarrow$ PWB	0.139	0.082	1.685	0.110	-0.024	0.291	0.019
SS $\rightarrow$ PWB	0.110	0.075	1.426	0.168	-0.054	0.241	0.012
ED $\rightarrow$ PWB	0.124	0.063	1.937	0.139	-0.037	0.238	0.015

Note: Std. = standardized; BCI = bias-corrected bootstrap confidence interval; WE = working environment; SS = social support; ED = economic domains; QoL = quality of life; PWB = psychological well-being.

Table 5. Hypothesis testing (indirect effect)

	Std. β	Std. Error	t -value	p -value	BCI Lower Limit	BCI Upper Limit
WE $\rightarrow$ QoL $\rightarrow$ PWB	0.101	0.025	4.069	0.000	0.052	0.150
SS $\rightarrow$ QoL $\rightarrow$ PWB	0.079	0.023	3.363	0.001	0.036	0.125
ED $\rightarrow$ QoL $\rightarrow$ PWB	0.282	0.024	11.477	0.000	0.234	0.330

Note: Std. = standardized; BCI = bias-corrected bootstrap confidence interval; WE = working environment; SS = social support; ED = economic domains; QoL = quality of life; PWB = psychological well-being.

The analysis focus on the effect of three variables on overall QoL. The results show that R^2 is 0.721 ($Q^2 = 0.507$), which indicates that the three variables explained 72.1% of the variance in overall QoL. Working environment ($\beta = 0.212$, $p < 0.05$), Social support ($\beta = 0.161$, $p < 0.05$), and economic domains ($\beta = 0.587$, $p < 0.05$) were shown to have a significant positive relationship with overall QoL. So, it can be concluded that H1, H2 and H3 were supported. However, the direct paths from working environment, social support, and economic domain to psychological well-being have also been tested to be non-significant, with p -value more than 0.05 and T -statistic value lower than 1.96 based on the study [49]. Thus, hypotheses H8, H9 and H10 were not supported. Besides that, overall QoL ($\beta = 0.477$, $p < 0.05$) was shown to have a significant positive relationship with psychological well-being. Hence, the results showed that H4 was supported.

Subsequently, hypotheses for mediation impact were tested, following the suggestions from the study [50] to bootstrap the indirect effect. If the confidence interval does not include zero then we can conclude that there is significant mediation. As shown in Table 5, WE $\rightarrow$ QoL $\rightarrow$ PWB ($\beta = 0.101$, $p < 0.05$), SS $\rightarrow$ QoL $\rightarrow$ PWB ($\beta = 0.079$, $p < 0.05$) and ED $\rightarrow$ QoL $\rightarrow$ PWB ($\beta = 0.282$, $p < 0.05$) were all significant. The bias-corrected 95% confidence intervals also did not include zero thus confirming our findings. Thus, H5, H6 and H7 were also supported.

5 Discussion

The analysis of the study supports hypotheses H1, H2, and H3. This demonstrates a strong positive correlation between WE, SS, ED, and overall QoL. H1 reveals that working environment conditions significantly influence overall QoL. This result can be explained by practical aspects such as ergonomic conditions and supportive managerial practices are crucial for enhancing truck drivers' life quality. These aspects are not merely soft human resource elements, but are critical determinants of truck driver's overall life quality [52]. Furthermore, this resonates with the findings [29], which shows that significant impact of ergonomic designs and supportive work policies would improve workforce satisfaction and job outcomes. Similarly, the study [30] also highlights the critical importance of favorable working conditions in promoting employee overall QoL. Thus, these studies reinforce the idea that

improving the working environment is a key determinant of life quality for building workforce resilience in the Malaysian context [53].

Next, H2 reveals that social support significantly influences overall QoL. It serves as a powerful mechanism in mitigating occupational stress and enhancing psychological health [32]. The study [54] highlights that workplace social support could significantly reduce burnout symptoms. This is because the nature of long-haul truck driving is structurally isolated from conventional office settings, peer and supervisor cooperation would serve as an informal buffer against operational fatigue and psychological burnout [55]. In fact, the lack of social support from supervisors and co-workers would lead to job insecurity that subsequently decreases the satisfaction of overall QoL for truck drivers [38]. Unlike highly regulated transport sectors in Western nations, a large portion of Malaysia's haulage is managed by small and medium enterprises (SMEs) where formal psychological health support systems and strict occupational driving-hour enforcement are severely restricted [9]. Thus, localized interpersonal support systems are crucial for truck drivers with isolated working environments.

The H3 findings show that economic domains concerned job security and income with reasonable cost of living are vital determinants of truck drivers' overall QoL. This is in line with the findings of the study [56], which underscore the role of economic factors in mitigating the burdens among truck drivers through improved job security. From an engineering management perspective, if truck drivers constantly work under highly volatile and trip-based pay models without base income stability, they are financially incentivized to compromise safety protocols that violate safety requirements by extending driving hours [57]. This behavior would introduce severe logistical risk that increases accident rates and causing catastrophic delivery disruptions [58]. Additionally, the study [59] supports that economic stability can mitigate job stress and enhance QoL through supportive working environments and organizational support.

Hypothesis H4 reveals a significant positive impact of overall QoL on the psychological well-being of truck drivers. This pivotal finding demonstrates that drivers' positive life growth and satisfaction perceptions would significantly contribute to their psychological health. The findings align with empirical evidence from prior studies [18, 20] that supported the improved life quality is closely associated with enhanced psychological states among individuals.

5.1 The Role of Overall Quality of Life

The structural model shows that the relationships between WE (H8), SS (H9), ED (H10) and PWB are statistically non-significant with p -value more than 0.05. This indicates an important theoretical distinction in occupational psychology, where external occupational factors would act as distal stressors [14]; whereas psychological well-being represents a deeply internal and intrinsic emotional state [60]. The working environment, social support and economic domains measure organizational and transactional inputs such as peer cooperation, management communication and income satisfaction. For structurally isolated truck drivers, these external occupational factors do not possess the contextual immediacy to directly alter internal emotional well-being [14]. For instance, transactional compensation does not directly generate eudaimonic happiness [61]. Thus, these occupational factors must permeate the truck driver's holistic evaluation of their daily existence [62]. Therefore, this further confirms H5, H6, H7, supporting the full mediating role of overall QoL between WE, SS, ED and the PWB of truck drivers. This is consistent with the findings of the study [3], where improved work conditions, sufficient social support, and economic stability are integral to fostering mental resilience and health. The truck driver represents a vulnerable 'human-in-the-loop' logistics operational node from engineering management perspective. Psychological well-being among truck drivers becomes a critical measurement of service delivery and operational reliability [63]. A psychologically supported and resilient workforce ensures consistent service delivery, stabilizes transport operations during peak demands, and ultimately optimizes the overall efficiency of the supply chain [64]. The study [59] explored psychological wellness among long-haul truck drivers. This highlights the critical role of work organization in affecting truck drivers' work-life balance [42]. The complex interplay between occupational factors and well-being generally provides an empirical foundation that echoes the emphasis on the overall QoL's influence on psychological well-being [38]. The results of the mediation analysis emphasize the integration of financial or environmental stressors as a whole. To further illustrate, the 'tension' underpinned by balance theory would reduce psychological resilience [20]. To regain balance, the truck driver requires positive interventions in the 'O' with social support or improved economic domains to restore a positive triad. This would further underline the pivotal role of psychological well-being among truck drivers [38]. Logistics managers should utilize QoL as a leading predictive indicator of workforce reliability. By auditing and adjusting the primary systemic inputs such as engineering ergonomic workspaces, formalizing peer-support communication channels, and stabilizing trip-based compensation models, logistics managers can help secure the truck drivers' baseline QoL. This, in turn, may help prevent the degradation of psychological well-being among drivers. By proactively improving the working environment and securing economic domains, logistics managers can foster a high degree of workforce resilience. Collectively, these would improve overall psychological well-being behind the wheel.

6 Conclusion

In conclusion, this research demonstrates that driver well-being is an indispensable pillar of transport system stability. It assists in bridging the gap between the field of occupational psychology and engineering management. The results reveal the significant insights into how QoL critically mediates the influence of working environment, social support, and economic domains on drivers' psychological well-being. This would enable decision-makers to mitigate operational safety risks and reduce turnover by treating the truck drivers' workforce as an integral living component of the logistics infrastructure. Holistic interventions targeting drivers' QoL will not only improve individual psychological health but also engineer more resilient workforces. Therefore, it would serve as an insightful perspective for industry practices and policymaking aimed at improving to improve truck drivers' well-being. Although this study provides broad insights into the collective psychological well-being of Malaysian truck drivers, the demographic distribution presents certain boundaries to its generalizability. Specifically, the sample features a relatively high proportion of 25.6% female drivers and a diverse mix of age groups and operational experience. This study did not isolate these demographic differences as workforce population is evaluated holistically. Future research should leverage multigroup structural equation modeling to explicitly compare differential psychological paths between male and female drivers or between less experiences and high experiences truck drivers' cohorts. Therefore, this would allow logistics managers to engineer more targeted and demographic-specific intervention frameworks for truck drivers. Furthermore, this research not only strengthens existing theories on occupational health but also presents novel insights into the operational health and safety context. This would further contribute to the body of knowledge in occupational psychology and the field of well-being research.

Author Contributions

Conceptualization, J.F.Y. and B.A.T.; methodology, J.F.Y.; software, J.F.Y.; validation, J.F.Y. and B.A.T.; formal analysis, B.A.T.; investigation, B.A.T.; resources, B.A.T.; data curation, B.A.T.; writing—original draft preparation, J.F.Y.; writing—review and editing, B.A.T.; visualization, B.A.T.; supervision, B.A.T.; project administration, B.A.T.; funding acquisition, B.A.T. All authors have read and agreed to the published version of the manuscript.

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Informed Consent Statement

Informed consent was obtained electronically from all individual participants prior to their engagement in the study. Before accessing the survey questionnaire, participants were presented with a digital information sheet detailing the study's objectives, the voluntary nature of their participation, and their right to withdraw at any time without penalty. Complete anonymity and data confidentiality were guaranteed, as no personal identifiers (such as names, phone numbers, or corporate emails) were recorded. Participants explicitly provided consent by checking a mandatory confirmation box to launch the survey.

Ethical Approval

Formal institutional ethical review and approval were waived for this study because the research design relied entirely on an anonymous, non-invasive online questionnaire that posed minimal risk to participants. No sensitive personal data, biological samples, or identifying medical records were collected. The study was conducted in strict accordance with the ethical principles of the Declaration of Helsinki, ensuring that data collection practices upheld safety, privacy, and integrity throughout the research process.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The author used generative AI tools solely for limited language editing and formatting assistance during manuscript preparation. The authors remain fully responsible for the content of the work and for ensuring its accuracy, originality, and compliance with ethical and publishing standards. Generative AI tools may not be listed as authors. The use of generative AI to fabricate data, results, references, or to substantially replace human intellectual contribution is not permitted.

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